APPENDICES A–F

Reference Companion

DevSecOps Lab Book · Quick Reference, Commands, Glossary, Templates, Troubleshooting & Resources

6
Phases Covered

150+
Commands Indexed

40+
Glossary Terms

Instructor reference set for hands-on labs, troubleshooting sessions, and self-paced study.

What's Inside

Six appendices supporting the DevSecOps Lab Book curriculum.



A

Quick Reference Cards

One-page cheat sheets for all 6 lab phases.



B

Command Reference

Essential CLI commands by category.



C

Glossary

40+ core DevSecOps terms & definitions.



D

Templates

Ready-to-use K8s, Terraform, Ansible & CI/CD configs.



E

Troubleshooting

Common problems, causes & solutions.



F

Learning Resources

Docs, books, communities & certification paths.

Phase 1: Foundation

BIND9 DNS Setup

Install: apt-get install bind9

Config: /etc/bind/named.conf

Split-brain: internal vs external zones

Test: dig @10.0.2.10 devsecops.com

LDAP / Authentik Identity

LDAP: ldap://10.0.2.11:389, base dc=devsecops,dc=com

OIDC: http://auth.devsecops.com:9000

Create user: ldapadd -D "cn=admin,dc=devsecops,dc=com"

Test: ldapsearch -x -H ldap://10.0.2.11

HAProxy Load Balancer

Config: /etc/haproxy/haproxy.cfg

Frontend: bind *:443, default_backend backend

Backend: balance roundrobin, server cp-01 10.0.2.20:6443

Test: curl -k https://10.0.2.12

Phase 1–2: Foundation & Automation

HTTPS / TLS Certificate

Generate: openssl req -x509 -newkey rsa:4096 ...

Verify: openssl x509 -in cert.pem -text -noout

Authority: /etc/ssl/certs/ca-certificates.crt

HAProxy: cat cert.key cert.crt > devsecops.com.pem

Packer Image Building

Template: ubuntu-22.04.pkr.hcl

Provisioners: provisioners.sh (install tools)

Build: packer build ubuntu-22.04.pkr.hcl

Output: .../ubuntu-22.04-packer.qcow2

Terraform IaC

Init / Plan: terraform init → terraform plan -out=tfplan

Apply: terraform apply tfplan

Destroy: terraform destroy

State: terraform show | terraform state list

Phase 2–3: Automation & Security

Ansible Playbooks

Inventory: inventory.ini (host list)

Playbook: site.yml (configuration)

Run: ansible-playbook -i inventory.ini site.yml

Check: ansible all -m ping

Vault Secrets

Init / Unseal: vault operator init → unseal (3x with keys)

Write / Read: vault kv put secret/myapp password=secret

Admin: vault audit enable file file_path=...

Teleport Access

Login: tsh login --proxy=teleport.devsecops.com:3023

SSH: tsh ssh ubuntu@cp-01

Record: sessions recorded in /var/lib/teleport

Audit: tctl sessions ls

Phase 3–4: Security & CI/CD

PKI Certificates

Generate: vault write pki_int/issue/lab common_name=...

Renew: vault write -f pki_int/rotate/crl

Revoke: vault write pki_int/revoke serial_number=xxx

GitLab Pipelines

Config: .gitlab-ci.yml

Stages: build → test → deploy

Access: https://gitlab.devsecops.com

Runner: gitlab-runner install & start

SonarQube SAST

Scan: sonar-scanner -Dsonar.projectKey=myapp

Dashboard: https://sonarqube.devsecops.com

Quality Gate: > 80% coverage, 0 critical issues

Phase 5: Advanced Operations

Harbor Registry

Login: docker login harbor.devsecops.com

Push: docker push harbor.devsecops.com/myapp:latest

Scan: trivy image harbor.devsecops.com/myapp:latest

Policy: Block images with critical vulnerabilities

High Availability

Vault HA: Consul backend, 3+ nodes

PostgreSQL: Patroni auto-failover

Redis: Sentinel with 3+ nodes

VIP: Keepalived at 10.0.2.100

Disaster Recovery

Backups: Hourly snapshots, 30-day retention

Test recovery: Monthly

RTO / RPO: < 1 hour / < 15 minutes

Phase 6: Kubernetes

Kubernetes Cluster

Init: kubeadm init --control-plane-endpoint=10.0.2.25:6443

Join Worker: kubeadm join --token xxx

CNI: kubectl apply -f calico.yaml

Verify: kubectl get nodes

Helm Charts

Create: helm create myapp

Install: helm install myapp ./myapp-chart

Upgrade / Rollback: helm upgrade myapp ./chart / helm rollback

GitOps (ArgoCD)

Install: kubectl apply -n argocd -f install.yaml

Create App: argocd app create myapp --repo https://...

Sync / Rollback: argocd app sync myapp / argocd app rollback myapp 2

System Admin & Kubernetes

System Administration

```
$ ping 10.0.2.10  # Test connectivity

$ netstat -tulpn  # Show listening ports

$ ip route show  # Show routing table

$ dig devsecops.com +trace  # Trace DNS

$ systemctl restart sshd  # Restart service
```

Kubernetes

```
$ kubectl cluster-info  # Cluster details

$ kubectl get pods -A  # All pods

$ kubectl logs pod-name -n ns  # Container logs

$ kubectl rollout undo deployment/app  # Rollback

$ kubectl explain deployment  # Resource docs
```

Security & Containers

Security

```
$ openssl x509 -in cert.pem -text -noout  # View certificate

$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519  # Generate key

$ ufw allow 22/tcp  # Allow SSH

$ ausearch -k passwd_changes  # Search audit logs

$ aureport --user-summary  # User summary
```

Containers

```
$ docker build -t myapp:latest .  # Build image

$ docker push harbor.devsecops.com/myapp:latest  # Push image

$ docker exec -it container-id /bin/bash  # Execute in container

$ trivy image harbor.devsecops.com/app:latest  # Scan image

$ trivy fs /path/to/filesystem  # Scan filesystem
```

Git & Ansible

Git

```
$ git clone https://url.git  # Clone repo

$ git checkout feature/new-feature  # Switch branch

$ git commit -m "message"  # Commit changes

$ git push origin main  # Push to remote

$ git pull origin main  # Pull from remote
```

Ansible

```
$ ansible all -i inventory.ini -m ping  # Test connectivity

$ ansible-playbook -i inventory.ini site.yml  # Run playbook

$ ansible-playbook site.yml --check  # Dry-run

$ ansible all -m shell -a "echo hello"  # Run command

$ ansible all -m service -a "name=ssh state=started"  # Service
```

APPENDIX C • GLOSSARY

Core Concepts & Infrastructure

DevSecOps
Integration of security into DevOps, shifting security left in the dev lifecycle.

Defense in Depth
Multiple security layers; if one fails, others remain.

Zero Trust
No default trust — all access requires authentication & authorization.

IaC
Define infrastructure using code (Terraform, Ansible).

Idempotent
Operation produces the same result no matter how many times it runs.

Immutable Infrastructure
Infrastructure is replaced rather than updated.

APPENDIX C • GLOSSARY

Kubernetes & Security

Control Plane
Kubernetes brain: API server, scheduler, etcd, controller manager.

Pod
Smallest Kubernetes unit; contains one or more containers.

RBAC
Role-Based Access Control; users are granted roles with permissions.

mTLS
Both client & server authenticate each other.

SAST / DAST
Static / Dynamic Application Security Testing.

Exploit
Code or technique that uses a vulnerability to compromise a system.

APPENDIX C • GLOSSARY

Compliance, Ops & Deployment

SOC 2 / ISO 27001
Security & operational controls frameworks.

CIS Benchmarks
Center for Internet Security best-practice standards.

RTO / RPO
Max acceptable downtime / max acceptable data loss.

MTTR
Mean Time To Repair — average time to fix issues.

Blue-Green
Two identical environments; switch traffic between them.

Canary
Deploy to a small user subset first, then roll out fully.

APPENDIX D · TEMPLATES

Kubernetes & Infrastructure as Code

K8s Pod Security Template

runAsNonRoot: true, runAsUser 1000, seccompProfile RuntimeDefault

Container context: allowPrivilegeEscalation: false, readOnly rootfs

Capabilities: drop ALL, add NET_BIND_SERVICE only

Resources: limits 500m/512Mi, requests 250m/256Mi

Probes: liveness /health, readiness /ready

Terraform Module Template

Variables: cluster_name, node_count (default 3), machine_type

Resource: kubernetes_cluster with node_pool block

Node config: machine_type + disk_size_gb: 100

Outputs: cluster_endpoint, cluster_ca_certificate

Use When

- Pod template: hardening any production workload
- Terraform module: standardizing cluster provisioning

APPENDIX D · TEMPLATES

Automation, Pipelines & Network Security

Ansible Playbook Template

Target: hosts: kubernetes_masters

Vars: app_name, app_version, container_registry

Tasks: create namespace → regcred secret → helm deploy

Verify: k8s_info until readyReplicas == spec.replicas

GitLab CI/CD Template

Stages: build → test → security → deploy

build: docker build & push to Harbor registry

security: sonar-scanner SAST + dependency-check

deploy: helm upgrade --install, only: main

Firewall Rules Template

Defaults: deny incoming, allow outgoing, deny routed

Essential: 22/tcp, 80/tcp, 443/tcp

Kubernetes: 6443/tcp (API), 10250/tcp (kubelet)

Hardening: ufw limit 22/tcp for brute-force protection

APPENDIX E • TROUBLESHOOTING

Kubernetes & Security Issues

Pods stuck in Pending

Insufficient resources, node affinity, PVC not bound, missing service account.

Check kubectl describe pod → free resources or fix scheduling constraints.

CrashLoopBackOff

App crashes, missing dependencies, or resource limits too low.

kubectl logs --previous, then raise limits or fix the underlying error.

Certificate errors (SSL/TLS)

Expired or mismatched certificate chain.

Regenerate via Vault PKI and update the Kubernetes TLS secret.

Authentication failures

Stale credentials or OIDC/Teleport session expired.

Refresh kubeconfig token and re-run tsh login.

APPENDIX E • TROUBLESHOOTING

Network & Performance Issues

Service not accessible

Selector doesn't match pod labels, or wrong container port.

Verify svc selector/ports and confirm endpoints are populated.

DNS not resolving

BIND not running, bad zone file, or ACL blocking query.

systemctl restart named, then named-checkzone to validate.

High CPU / memory usage

Under-provisioned limits or an application memory leak.

kubectl top pods, raise requests/limits, then scale horizontally.

Slow application response

Missing connection pooling or cold caches.

Add pgBouncer pooling and Redis caching; load-test with JMeter.

APPENDIX F · LEARNING RESOURCES

Documentation & Certifications

Official Docs

- kubernetes.io/docs — core & security concepts
- docs.docker.com — engine & best practices
- registry.terraform.io & docs.ansible.com
- cisecurity.org — CIS Benchmarks

Certifications

- CKA — Certified Kubernetes Administrator
- CKAD — Certified K8s App Developer
- CISSP — Information Systems Security
- AWS / Azure Security Engineer tracks

Recommended Books

- "Kubernetes in Action" — Marko Lukša
- "The Phoenix Project" — Gene Kim
- "Zero Trust Networks" — Gilman & Barth
- "Accelerate" — Nicole Forsgren

APPENDIX F • LEARNING RESOURCES

Communities & Tooling

Communities

- Kubernetes Slack & CNCF Community
- DevSecOps Community forums
- Cloud Native Security Community
- Local Kubernetes / DevOps meetups

Container & K8s Security

- Trivy — vulnerability scanner
- Cosign — image signing
- Kyverno / OPA Gatekeeper — policy
- Kube-bench — CIS benchmarks

Monitoring & IR

- Prometheus + Grafana — metrics
- ELK Stack — logging, Jaeger — tracing
- osquery / auditd — system monitoring
- TheHive / MISP — incident response

APPENDIX F • LEARNING RESOURCES

Continuing Education Plan

Monthly Rhythm

- Week 1 — read one paper or blog post
- Week 2 — complete one lab exercise
- Week 3 — practice with the tools
- Week 4 — review and consolidate

Annual Plan

- Q1 — Fundamentals: OS & networking
- Q2 — Infrastructure: K8s & IaC
- Q3 — Security: hardening & IR
- Q4 — Advanced: multi-region & compliance

Stay Current

- Follow CISA alerts & vuln disclosures
- Subscribe to DevSecOps newsletters
- Contribute to open source projects
- Attend conferences, meetups & webinars

Using This Companion

- Hand out Appendix A cards at the start of each lab phase as a one-page cheat sheet.
- Keep Appendix B open in a side window during hands-on exercises for copy-paste commands.
- Use Appendix C to pre-teach vocabulary before introducing a new phase's concepts.
- Adapt Appendix D templates directly into student lab environments.
- Run Appendix E scenarios as live debugging drills; Appendix F guides self-paced study after class.